

Experiment No: 13

Title

Write a Python Program to Implement Minimax Algorithm for Gaming

Aim

To develop a Python program to implement the **Minimax Algorithm** for selecting the optimal move in a two-player game.

Objective

To implement the Minimax algorithm to evaluate game states and determine the best possible move for a player.

Algorithm

Step 1: Start the program.

Step 2: Define the game tree values.

Step 3: Check whether the current node is a leaf node.

Step 4: If it is the maximizing player's turn, select the maximum value.

Step 5: If it is the minimizing player's turn, select the minimum value.

Step 6: Repeat recursively until the root node is evaluated.

Step 7: Display the optimal value.

Step 8: Stop the program.

Program

```
# Minimax Algorithm
```

```
scores = [3, 5, 2, 9, 12, 5, 23, 23]
```

```
def minimax(depth, node, maximizing):
```

```
    if depth == 3:
```

```

        return scores[node]

    if maximizing:

        left = minimax(depth + 1, node * 2, False)
        right = minimax(depth + 1, node * 2 + 1, False)

        return max(left, right)

    else:

        left = minimax(depth + 1, node * 2, True)
        right = minimax(depth + 1, node * 2 + 1, True)

        return min(left, right)

result = minimax(0, 0, True)

print("Optimal Value :", result)

```

Sample Input

No user input.

Sample Output

```

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Optimal Value : 12
|
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```

Result

Thus, the Python program to implement the **Minimax Algorithm** was executed successfully, and the optimal game value was obtained correctly.

